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Karyotype Analysis of Four *Vicia* Species using *In Situ* Hybridization with Repetitive Sequences

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Mitotic chromosomes of four *Vicia* species (*V. sativa*, *V. grandiflora*, *V. pannonica* and *V. narbonensis*) were subjected to *in situ* hybridization with probes derived from conserved plant repetitive DNA sequences (18S–25S and 5S rDNA, telomeres) and genus-specific satellite repeats (VicTR-A and VicTR-B). Numbers and positions of hybridization signals provided cytogenetic landmarks suitable for unambiguous identification of all chromosomes, and establishment of the karyotypes. The VicTR-A and -B sequences, in particular, produced highly informative banding patterns that alone were sufficient for discrimination of all chromosomes. However, these patterns were not conserved among species and thus could not be employed for identification of homologous chromosomes. This fact, together with observed variations in positions and numbers of rDNA loci, suggests considerable divergence between karyotypes of the species studied. © 2003 Annals of Botany Company

Key words: *Vicia*, *in situ* hybridization, PRINS, rRNA genes, satellite DNA, repetitive DNA, karyotype evolution.

INTRODUCTION

The genus *Vicia* includes over 160 species with a primarily Euro-Asiatic distribution (Kupicha, 1976; Allkin *et al*., 1986; Maxted, 1995), some of which, such as *V. faba* (field bean), *V. narbonensis* (narbon vetch) and *V. sativa* (common vetch), are economically important crops. There is a considerable variation in haploid nuclear DNA content (1·8–13·3 pg) and basic chromosome number ($2n = 10$, 12 or 14) between *Vicia* species (Raina and Narayan, 1984; Maxted, 1995; Bennett and Leitch, 1998), which makes the genus an interesting model for the study of plant genome and karyotype evolution. The karyotypes of several species have been established based on chromosome size, centromeric index and banding patterns (Cremonini, 1992; Galasso *et al*., 1994; Cremonini *et al*., 1998; Fuchs *et al*., 1998); however, this approach is limited by the similar morphology of chromosomes in many species. Visualization of specific DNA sequences on chromosomes using FISH (fluorescence *in situ* hybridization) or PRINS (primed *in situ* DNA labelling) can be used to overcome these limitations by providing specific labelling patterns useful for discrimination of similar chromosomes. Probes for FISH or PRINS can be derived from conserved repetitive sequences, such as 18S–25S (18S–5.8S–25S) rRNA or 5S rRNA genes, which have been used successfully for chromosome discrimination in several *Vicia* species (Galasso *et al*., 1997; Raina *et al*., 2001). Additional cytogenetic landmarks can be obtained using species- or genus-specific satellite repeats that are often amplified to high copy numbers and form discrete bands or spots on chromosomes (Charlesworth *et al*., 1994; Schmidt and Heslop-Harrison, 1998). Compared with rRNA genes, these repeats frequently occur at a higher number of genomic loci and may therefore produce signals characteristic for each chromosome within the karyotype, as was shown for *V. faba* (Fuchs *et al*., 1994, 1998) and *V. sativa* (Nouzová *et al*., 2001).

In our previous work (Macas *et al*., 2000) two families of satellite repeats, VicTR-A and VicTR-B, have been isolated. These are specific for the genus *Vicia* and show different levels of amplification in individual species. The copy numbers of both repeats range from undetectable to up to 10^6 per haploid genome; however, VicTR-A repeats are amplified mainly in species from the taxonomic section Hypechusa and in *V. narbonensis* (section Narbonensis), whereas VicTR-B sequences are most abundant in section Vicia (which lacks VicTR-A repeats) and, interestingly, also in *V. narbonensis.* The aim of the present study was to use VicTR sequences in combination with conserved plant repeats (18S–25S and 5S rRNA genes) for FISH and PRINS labelling of chromosomes in four *Vicia* species differing in genome size and chromosome numbers (Table 1). The main objectives were: (1) to discriminate individual chromosome types and to establish karyotypes in these species; (2) to investigate whether the distribution patterns of VicTR repeats are conserved among the species; and (3) to assess the possibility of identifying homeologous chromosomes based on these hybridization patterns.

MATERIALS AND METHODS

Plant material, cell cycle synchronization and chromosome preparation

Seeds of *Vicia* species were obtained from the germplasm collection of the Institute of Plant Genetics and Crop Plant Research, Gatersleben, Germany (*V. grandiflora* Scop.), Dr

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TABLE 1. *Chromosome numbers, DNA content and abundance of VicTR repeats in selected* Vicia *species*

Section*	Species	Chromasome number ($2n$)	DNA content (pg)	Abundance of VicTR repeats	
				VicTR-A	VicTR-B
Vicia	*V. sativa*	12	2·3	–	10^6–5×10^6
	V. grandiflora	14	3·4	100–1000	10^6–5×10^6
Hypechusa	*V. pannonica*	12	6·8	10^5–10^6	~100
Narbonensis	*V. narbonensis*	14	7·3	10^5–10^6	10^5–10^6

* Assignment of species to taxonomic sections is according to Kupicha (1976) and Maxted (1995).
† Number of copies per haploid genome, according to Macas *et al.* (2000).

I. Nianiou, Aristotelian University of Thessaloniki, Greece (*V. narbonensis* L. 'IFYN574'), and from plant breeding stations at Horní Moštěnice (*V. sativa* L. 'Ebena') and Chlumec nad Cidlinou, Czech Republic (*V. pannonica* Crantz 'Dětenická panonská'). Three-day-old seedlings (*V. sativa*, *V. pannonica* and *V. narbonensis*) or hairy root cultures (*V. grandiflora*) were used for root meristem synchronization and accumulation of cells in metaphase as described by Doležel *et al*. (1992) and Neumann *et al*. (1998), respectively, with the following modifications: hydroxyurea (ICN Biomedicals, Irvine, CA, USA) treatment was carried out for 18 h at 1 mM concentration for *V. sativa*, 0·75 mM for *V. pannonica*, 1·25 mM for *V. narbonensis* and 1·5 mM for *V. grandiflora*; the time in hydroxyurea-free medium was 2 h (*V. sativa*, *V. pannonica*), 4·5 h (*V. narbonensis*) or 6 h (*V. grandiflora*); and metaphase accumulation was achieved using 15 μM oryzalin (Chem Service, West Chester, PA, USA) applied for 7 h (*V. narbonensis*) or 4 h (all other species). Synchronized root tips were used for chromosome preparations using a squashing or dropping method as described by Leitch *et al*. (1994) and Busch *et al*. (1996). Alternatively, the chromosomes of *V. sativa*, *V. pannonica* and *V. narbonensis* were prepared as purified chromosome suspensions (Gualberti *et al*., 1996) and centrifuged onto microscope slides using a Hettich Universal 32-R centrifuge equipped with cytospin chambers.

FISH and PRINS

The different methods of chromosome preparation described above and direct or indirect probe detection systems were employed so that optimal sensitivity could be achieved and the results could be verified using alternative methods. Probes for FISH were labelled with either fluorochrome- (Alexa Fluor 488-dUTP or Alexa Fluor 568-dUTP, Molecular Probes, Eugene, OR, USA) or biotin-conjugated dUTP (biotin-16-dUTP; Boehringer-Mannheim, Mannheim, Germany). Labelling of VicTR and 5S rDNA repeats (the 5S rDNA was cloned from pea; Neumann *et al*., 2001) was performed in a PCR reaction using universal primers and corresponding sequences cloned into a plasmid vector as a template. Alternatively, the VicTR-A and -B sequences were labelled by direct amplification from genomic DNA of *V. pannonica* and *V. sativa* as described by Macas *et al*. (2000). The probe for 18S–25S rDNA was prepared by random priming of the clone Ver17 (Yakura and Tanifuji, 1983). Telomeric repeats were labelled using PCR without template DNA but including the complementary primers $(AGGGTTT)_3$ and $(CCCTAAA)_3$.

FISH experiments were performed as described by Neumann *et al*. (2001). Hybridization and washing temperatures were 37 °C and 42 °C, respectively, for rDNA and telomeric probes, and 28 °C and 32 °C for VicTR sequences. Simultaneous detection of two probes with different melting temperatures was done using a two-step hybridization protocol. The PRINS reactions were performed using pairs of sequence-specific primers and conditions described by Macas *et al*. (2000). For detection of two different targets the PRINS labelling was combined with FISH (PRINS–FISH technique). In these cases, the PRINS reaction was stopped by washing slides for 1·5 min in Stop buffer (50 mM NaCl, 50 mM EDTA, pH 8·0) at room temperature. The slides were then passed through an ice-cold ethanol series (50, 70, 96 %), air-dried and immediately used for FISH with omission of RNase pre-treatment, post-fixation step and slide denaturation. When PRINS was used for detection of VicTR repeats, subsequent FISH was performed at 28 °C to avoid washing out the VicTR signals.

Detection of biotin-labelled probes was done using streptavidin-fluorescein (Leitch *et al*., 1994) or using the Tyramide Signal Amplification Indirect system (NEN Life Sciences Products, Boston, MA, USA) with streptavidin-fluorescein at the final step. Probes labelled with Alexa-dUTP were visualized directly. Chromosomes were counterstained with DAPI (4′,6-diamidino-2-phenylindole) and observed using a Nikon Eclipse-600 epifluorescence microscope equipped with a CCD camera. Signals were collected separately using UV-2A, B-2A and G-2A filter sets, and processed using LUCIA software (Laboratory Imaging, Prague, Czech Republic).

Establishment of the karyotypes

Chromosome measurements were made using slides prepared by the squashing or dropping methods and subjected to FISH or PRINS with appropriate probe(s). At least eight complete metaphase figures with clearly distinguishable chromosomes were measured using LUCIA software, and average values of relative sizes of the chromosomes and their arms were used to construct

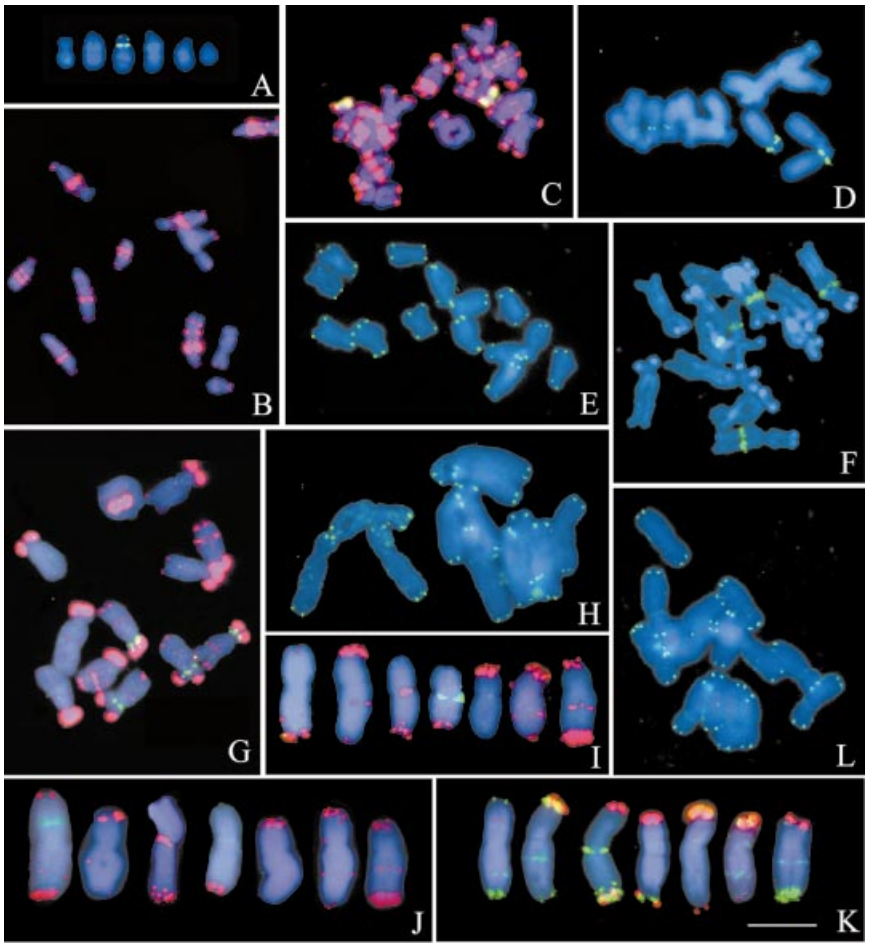


FIG. 1. Localization of repetitive sequences on metaphase chromosomes of *Vicia sativa* (A, B), *V. grandiflora* (C–E), *V. pannonica* (F–H) and *V. narbonensis* (I–L). 5S rDNA (A) and VicTR-B (B) sequences on *V. sativa* chromosomes. VicTR-B (red) and 18S–25S rDNA (green) (C), 5S rDNA (D), and telomeric probes (E) on *V. grandiflora*. 18S–25S rDNA (F), VicTR-A (red) and 5S rDNA (green) (G), and telomeric sequences (H) on *V. pannonica.* VicTR-B (red) and 18S–25S rDNA (green) (I), VicTR-B (red) and 5S rDNA (green) (J), VicTR-A (red) and VicTR-B (green) (K), and telomeric repeats (L) on *V. narbonensis* chromosomes. Sequences were localized using FISH except for VicTR repeats in B, C and G, which were visualized using PRINS. Chromosomes were counterstained with DAPI (blue). Bar = 10 μm.

idiograms. The rules for numbering chromosomes within idiograms generally followed those used by Hollings and Stace (1974).

RESULTS AND DISCUSSION

Vicia sativa

Vicia sativa has the smallest genome among the species investigated in this study (2·3 pg/1C). It possesses six pairs of chromosomes (one metacentric, four subacrocentrics and one acrocentric), two of which bear 18S–25S rRNA genes. The genes for 5S rRNA were detected at two pericentromeric loci on chromosome 3 (Fig. 1A). In *V. sativa*, only one of the VicTR repeats, VicTR-B, is amplified, and is present in large clusters with intercalary and (sub)terminal localization (Fig. 1B). VicTR-B signals corresponded to most of the DAPI-positive bands visible on mitotic chromosomes and produced patterns specific for each chromosome.

Vicia grandiflora

Although *V. grandiflora* belongs to the same taxonomic section as *V. sativa* (Kupicha, 1976; Maxted, 1995), it is divergent in chromosome number as well as in location of rDNA and VicTR-B sequences. Its genome consists of seven chromosome pairs (one metacentric, three submetacentrics and three subacrocentrics). Only one chromosome pair (chromosome 2) possesses a secondary constriction and gave a signal with the 18S–25S rDNA probe (Fig. 1C). 5S rRNA genes were detected at one locus on each of chromosomes 1 and 5 (Fig. 1D). As in *V. sativa*, the majority of DAPI bands showed hybridization with VicTR-B probe; however, these bands were located more at chromosome termini and were less intense within the chromosome arms (Fig. 1C).

Vicia pannonica

The six chromosome pairs of this species include four submetacentrics and two subacrocentrics, similar in size

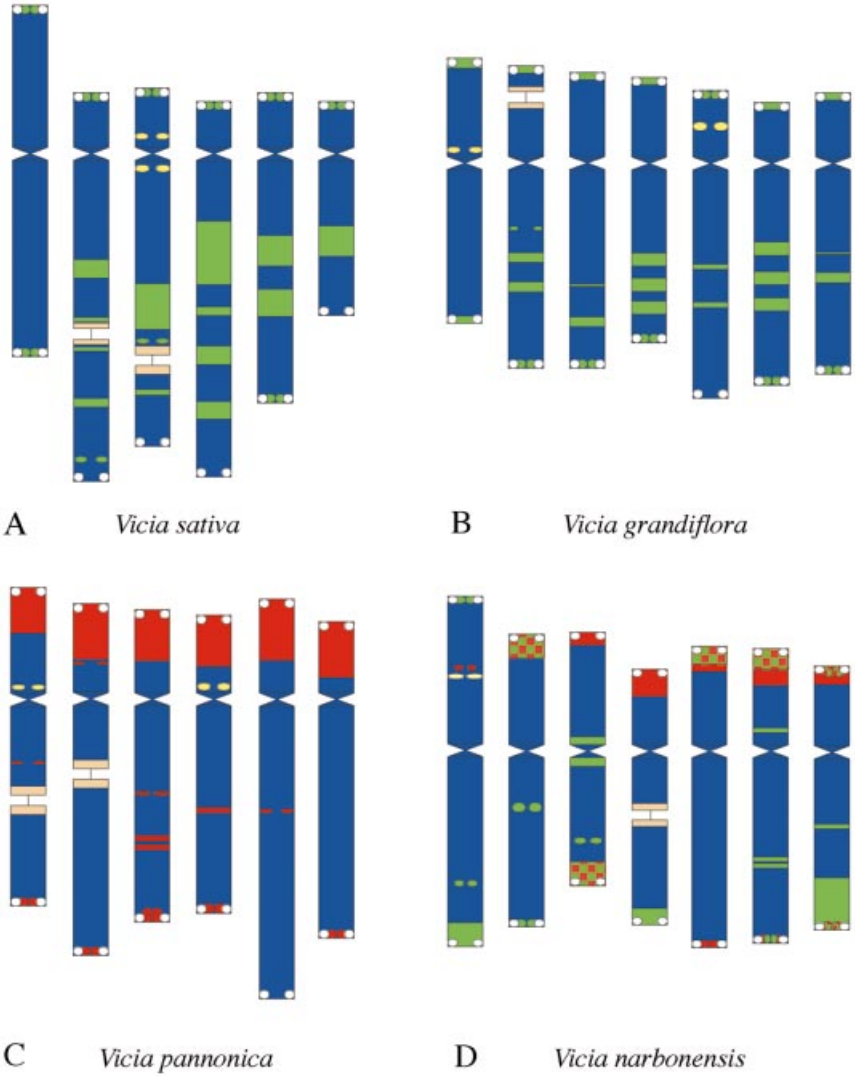


FIG. 2. Idiograms of *Vicia sativa* (A), *V. grandiflora* (B), *V. pannonica* (C) and *V. narbonensis* (D). Presence and distribution of 18S–25S rRNA genes (orange), 5S rRNA genes (yellow), VicTR-A (red), VicTR-B (green) and telomeres (white) is indicated.

and DAPI banding. Hybridization sites of 18S–25S rDNA probe corresponded to positions of NOR regions on chromosomes 1 and 2 (Fig. 1F), and 5S rDNA was observed on chromosomes 1 and 4 (Fig. 1G). Strong hybridization signals of VicTR-A sequences were found at the terminal regions of all chromosomes, and additional minor intercalary bands were visible on all chromosomes except chromosome 6 (Fig. 1G). Similarly to VicTR repeats in other species, the signals of VicTR-A were located at heterochromatic regions detectable after DAPI staining.

Vicia narbonensis

Vicia narbonensis represents a species with a relatively large genome (7·3 pg/1C) and is unique in its abundance of both families of VicTR repeats. Its complement consists of three metacentric and four submetacentric chromosomes, bearing one NOR on chromosome 4 (Fig. 1I) and one locus of 5S rDNA on chromosome 1 (Fig. 1J). VicTR-A repeats were located in the distal regions of all chromosomes except chromosome 1 (Fig. 1K); only a minor signal was observed on this chromosome and, in contrast to all other VicTR-A signals, it was located in an intercalary position within its short arm. A large number of VicTR-B sequences showed a similar distribution to those of VicTR-A, and the two repeats co-localized at the (sub-) telomeric regions of several chromosomes (Fig. 1K). However, the termini of some chromosomes hybridized to only one of these probes, and additional VicTR-B signals were observed as bands with intercalary and submetacentric locations. The preferential distribution of VicTR repeats at chromosome termini corresponds with the positions of heterochromatic regions detected by Cremonini *et al.* (1998) using C-banding and Hoechst 33258 staining.

TABLE 2. *Relative lengths and centromeric indexes of chromosomes*

	Vicia sativa		*Vicia grandiflora*		*Vicia pannonica*		*Vicia narbonensis*	
Arm*	Relative length ± s.d.†	Centromeric index ± s.d.‡	Relative length ± s.d.†	Centromeric index ± s.d.‡	Relative length ± s.d.†	Centromeric index ± s.d.‡	Relative length ± s.d.†	Centromeric index ± s.d.‡
1p	7·49 ± 0·63		5·17 ± 0·33		4·66 ± 0·27		7·62 ± 0·59	
q	10·31 ± 0·43	1·39 ± 0·15	8·10 ± 0·40	1·58 ± 0·16	12·54 ± 0·62	2·70 ± 0·17	9·93 ± 0·77	1·28 ± 0·11
2p	2·87 ± 0·48		4·23 ± 0·52		5·59 ± 0·55		5·79 ± 0·54	
q	16·40 ± 1·01	5·90 ± 1·19	10·43 ± 0·63	2·50 ± 0·36	9·99 ± 0·57	1·80 ± 0·19	8·87 ± 0·73	1·53 ± 0·20
3p	3·08 ± 0·39		4·42 ± 0·33		4·34 ± 0·52		5·59 ± 0·45	
q	14·46 ± 0·88	4·78 ± 0·82	10·47 ± 0·80	2·39 ± 0·35	11·42 ± 0·90	2·65 ± 0·31	6·60 ± 0·55	1·18 ± 0·13
4p	2·41 ± 0·47		4·17 ± 0·45		4·06 ± 0·53		3·95 ± 0·71	
q	16·69 ± 1·14	7·17 ± 1·51	9·07 ± 0·55	2·20 ± 0·32	10·95 ± 0·80	2·72 ± 0·30	8·25 ± 0·84	2·09 ± 0·23
5p	2·91 ± 0·45		3·52 ± 0·48		4·99 ± 0·52		5·12 ± 0·50	
q	12·78 ± 0·62	4·51 ± 0·83	12·02 ± 0·49	3·47 ± 0·50	15·42 ± 0·94	3·11 ± 0·31	9·94 ± 0·62	1·94 ± 0·17
6p	2·41 ± 0·38		2·92 ± 0·26		3·77 ± 0·19		5·01 ± 0·49	
q	8·17 ± 0·56	3·48 ± 0·69	11·33 ± 0·52	3·91 ± 0·41	12·27 ± 0·76	3·26 ± 0·29	9·70 ± 0·66	1·94 ± 0·22
7p			3·43 ± 0·35				4·17 ± 0·48	
q			10·73 ± 0·41	3·16 ± 0·38			8·96 ± 0·87	2·15 ± 0·28

* Short (p) and long (q) chromosome arms.
† Average (± s.d.) relative lengths in % of haploid genome with standard deviations.
‡ Centromeric indexes calculated as chromosome arm ratios (q/p) and supplied with standard deviations.

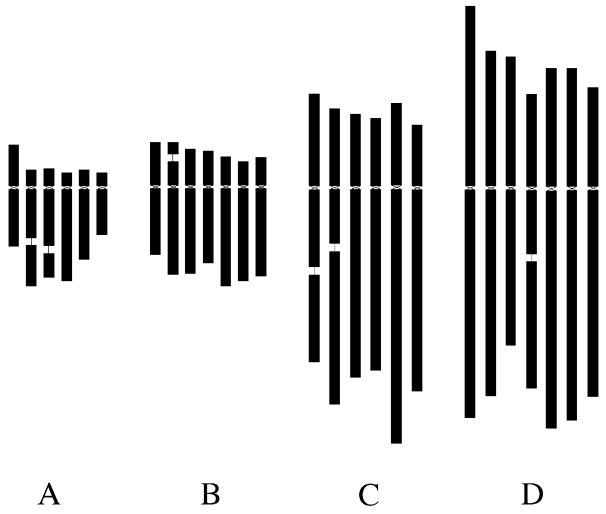


FIG. 3. Schematic comparison of chromosome sizes among *Vicia sativa* (A), *V. grandiflora* (B), *V. pannonica* (C) and *V. narbonensis* (D). Relative chromosome lengths were derived from the data in Table 2 and scaled according to the nuclear DNA content of the species (Table 1).

Establishment of karyotypes and their comparative analysis

The probes for VicTR repeats and rDNA sequences produced signals which allowed for the construction of karyotypes in all four species (Fig. 2). As all chromosomes were clearly discriminated, their relative sizes could be determined based on measurements of chromosome lengths of mitotic chromosomes subjected to FISH or PRINS with suitable probes (Table 2; Fig. 3). Establishment of the karyotypes also allowed us to address the question of identification of homologous chromosomes among the species. The hybridization patterns obtained after FISH with repetitive probes have been successfully used for this task in several plant genera (Lubaretz *et al*., 1996; Cuadrado and Jouve, 1997; Lim *et al*., 2000; Hizume *et al*., 2002). However, contrary to these reports, no homologous chromosomes could be unambiguously identified between the four *Vicia* species investigated here. Even the species from the same taxonomic section (*V. sativa* and *V. grandiflora*) displayed significant differences in chromosome morphology as well as in numbers and positions of hybridization signals (Fig. 2). This may be a consequence of a higher rate of karyotype evolution or a greater evolutionary distance among the species under study. The results of a comparative analysis of repetitive sequences (Nouzová *et al*., 2001) and morphological and isozyme diversity (Leht and Jaaska, 2002) indicate that the latter explanation is more likely.

Differential amplification of repetitive sequences may cause progressive changes of karyotypes in related species (Uozu *et al*., 1997; De la Herrán *et al*., 2001). Whereas amplification of retroelement sequences, which are generally dispersed in the genome (Pearce *et al*., 1996; Neumann *et al*., 2001), is likely to increase the size of all chromosomes within the karyotype in an approximately equal manner (Uozu *et al*., 1997), amplification of satellite DNA localized in large clusters has greater potential for increasing the asymmetry of the karyotypes (Uozu *et al*., 1997; De la Herrán *et al*., 2001). Moreover, satellite repeats are known to undergo rapid changes in copy numbers, resulting in their accumulation in a single species or a genus (Macas *et al*., 2002). As the copy numbers of VicTR-A and -B sequences vary by over six orders of magnitude (Macas *et al*., 2000), it can be supposed that this variability may have had an impact on differentiation of karyotypes in *Vicia*. A comparison of chromosome location of VicTR repeats in individual species revealed that the VicTR-B family is more variable compared with the VicTR-A family, which occurs almost exclusively at terminal regions of chromosome arms. It is interesting to note that on a level of

nucleotide sequences of the repeat monomers, VicTR-B appears to be more conserved than VicTR-A (Macas *et al*., 2000). However, more sequence data from different species are needed to confirm these findings and to study any possible correlation between sequence variability and genomic localization of these repeats.

As the majority of chromosomes displayed hybridization signals of VicTR probes at the termini of at least one of their arms, we investigated the presence of the *Arabidopsis*-type telomeric repeats (Richards and Ausubel, 1988) at these regions. We also searched for additional hybridization signals in intercalary or centromeric regions that could point to chromosome rearrangements. In contrast to *Vicia faba*, in which such interstitial signals were located on several chromosomes (Schubert, 1992), the telomeric sequences in all *Vicia* species tested in this work were detected only at terminal regions of their chromosomes (Fig. 1E, H, L).

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